Capstone Part 1 — Problem, Dataset, and Proposal

Start Assignment

- Due Thursday by 11:59pm
- Points 40
- Submitting a file upload
- Available Sep 14 at 12am - Oct 22 at 11:59pm

Part 1 establishes the foundation of your project. By the end of it, you should have a question worth answering, a dataset capable of answering it, and a realistic plan. Choosing well here determines how the rest of the semester goes.

What You Need to Do

1. Define your applied problem or research question

Answer clearly:

- What problem are you solving?
- Why does it matter?
- Who could benefit?
- What is the model input?
- What is the model output?

Frame this as a question, not a task. Write:

> Can deep learning models accurately identify flooded roads and buildings from aerial imagery?

rather than:

> I will train a CNN.

2. Describe your dataset

Report all of the following:

- Dataset name
- Dataset source
- Dataset link
- Dataset size
- Number of observations / images / documents
- Target variable
- Number of classes, if applicable
- Class distribution
- Input format

- Representative examples
- Known limitations
- License or usage requirements, if applicable

If your dataset was introduced in an academic paper, cite that paper in APA 7th Edition.

3. Conduct preliminary data exploration

Investigate and report on:

- Class imbalance
- Missing values
- Corrupted observations
- Image dimensions
- Text length
- Preprocessing needs
- Unusual observations

4. Propose your modeling approach

Briefly identify:

- Your planned baseline model
- Your proposed advanced model
- Approximately 2–3 experiments you may conduct

Your plans may change as the semester progresses. That is expected.

5. Verify GPU access

Demonstrate in your Kaggle Notebook that you have GPU acceleration:

python

```
import tensorflow as tf
print(tf.config.list_physical_devices('GPU'))
```

What to Submit

1. A short proposal/report covering items 1–4 above, formatted in APA 7th Edition
2. Your initial Kaggle Notebook, or a link to it

Checklist Before Submission

- My research question is phrased as a question, not a task
- I have stated the model input and the model output
- All eleven dataset details are reported
- My dataset link works and the dataset is publicly accessible
- I have reported class distribution, not just the number of classes

- I have described at least one real data problem I found during exploration
- I have named a baseline model and an advanced model
- I have listed 2–3 planned experiments
- My notebook shows a GPU is available
- If my dataset came from a paper, I have cited it in APA format
- My notebook link is shareable and does not require an access request